

```

use triggers;
-- Triggers:-
-- Triggers is a stored program that automatically fires when a
-- specific event (insert, update, delete) happens on a table.
-- update, delete, insert

-- before and after
-- NEW AND OLD
-- syntax
/*
DELIMITER $$
CREATE TRIGGER TRIGGER_NAME
[BEFORE/AFTER] [INSERT/UPDATE/DELETE]
ON TABLE_NAME
FOR EACH ROW
BEGIN
    SQL STATEMENTS
END $$
DELIMITER ;
*/

```

```

-- before insert trigger:-
-- runs before inserting a new record to your table.
select * from employees;

```

```

DELIMITER $$
CREATE TRIGGER BEFORE_EMPLOYEE_INSERT
BEFORE INSERT
ON EMPLOYEES
FOR EACH ROW
BEGIN
    SET NEW.SALARY = NEW.HOURLY_PAY * 2080;
END $$
DELIMITER ;

```

```

SHOW TRIGGERS;

```

```

INSERT INTO EMPLOYEES (FIRST_NAME, LAST_NAME, HOURLY_PAY, DOJ)
VALUES('SHREE', 'NATH', 25, '2022-09-08');

```

```

SELECT * FROM EMPLOYEES;

```

```

DELIMITER $$
CREATE TRIGGER BEFORE_INSERT_EMPLOYEES
BEFORE INSERT
ON EMPLOYEES
FOR EACH ROW

```

```

BEGIN
    set new.salary = new.hourly_pay * 2080;
END $$
DELIMITER ;

-- BEFORE UPDATE TRIGGER:-
delimiter $$
create trigger before_employee_update
before update
on employees
for each row
begin
    if new.salary < 0 then
        signal sqlstate '45000'
        set message_text = "salary can't be negative";
    end if;
end $$
delimiter ;

select * from employees;

update employees set salary = 80000
where e_id = 20;

-- before delete trigger:-
select *,date_sub(curdate(), interval 5 year) from employees;

delimiter $$
create trigger before_delete_employee
before delete
on employees
for each row
begin
    if old.doj < date_sub(curdate(), interval 5 year) then
        signal sqlstate '45000'
        set message_text = "cannot delete a senion employee";
    end if;
end $$
delimiter ;

select * from employees;
delete from employees where e_id = 6;

-- sql signal state
-- in mysql stored procedures and triggers, signal sqlstate '45000' is
-- used to manually raise a custom error.
-- it allows you to stop execution and return an error message when

```

```
-- condition is violated
-- syntax
-- signal sqlstate = '45000'
-- set message_text = 'your error message';

-- generic user defined error code --> 45000
```

```
-- 42000 --> syntax error
-- 23000 ---> constraints violation
-- 02000 --> no data found
```

```
-- After triggers:-
```

```
CREATE TABLE EMPLOYEES_LOGS(
EMP_LOG_ID INT PRIMARY KEY auto_increment,
EMP_ID INT,
EMP_FULL_NAME VARCHAR(225),
EMP_SALARY DECIMAL(9,2),
DOJ DATE,
CREATED_AT TIMESTAMP DEFAULT current_timestamp
);
```

```
-- AFTER INSERT
DELIMITER $$
CREATE TRIGGER AFTER_EMPLOYEE_INSERT
AFTER INSERT
ON EMPLOYEES
FOR EACH ROW
BEGIN
    INSERT INTO EMPLOYEES_LOGS(EMP_ID,EMP_FULL_NAME,
    EMP_SALARY, DOJ)
    VALUES (NEW.E_ID,CONCAT(NEW.FIRST_NAME, " ",NEW.LAST_NAME),
    NEW.SALARY,NEW.DOJ);
END $$
DELIMITER ;
```

```
-- AFTER UPDATE:-
```

```
SELECT * FROM EMPLOYEES;

DELIMITER $$
CREATE TRIGGER BEFORE_SALARY_UPDATE
BEFORE UPDATE
ON EMPLOYEES
```

```
FOR EACH ROW
BEGIN
    SET NEW.SALARY = NEW.HOURLY_PAY * 2080;
END $$
DELIMITER ;
```

```
CREATE TABLE SALARY_LOGS(
    SALARY_LOG_ID INT PRIMARY KEY auto_increment,
    EMP_ID INT,
    OLD_SALARY DECIMAL(9,2),
    NEW_SALARY DECIMAL(9,2)
);
```

```
DROP TABLE SALARY_LOGS;
SELECT * FROM EMPLOYEES;
```

```
DELIMITER $$
CREATE TRIGGER AFTER_SALARY_UPDATE
AFTER UPDATE
ON EMPLOYEES
FOR EACH ROW
BEGIN
    INSERT INTO SALARY_LOGS
        (EMP_ID, OLD_SALARY, NEW_SALARY)
    VALUES(OLD.E_ID, OLD.SALARY, NEW.SALARY);
END $$
DELIMITER ;
```

```
SELECT * FROM SALARY_LOGS;
```

```
-- AFTER DELETE:-
```

```
CREATE TABLE OLD_EMPLOYEES(
    EMP_ID INT PRIMARY KEY,
    EMP_FIRST_NAME VARCHAR(225),
    EMP_LAST_NAME VARCHAR(225)
);
```

```
DELIMITER $$
CREATE TRIGGER AFTER_DELETE_EMPLOYEES
AFTER DELETE
ON EMPLOYEES
FOR EACH ROW
BEGIN
    INSERT INTO OLD_EMPLOYEES
        (EMP_ID, EMP_FIRST_NAME, EMP_LAST_NAME)
```

```
VALUES (OLD.E_ID, OLD.FIRST_NAME, OLD.LAST_NAME);  
END $$  
DELIMITER ;
```

```
SELECT *, timestampdiff(YEAR, DOJ, CURDATE()) FROM EMPLOYEES;  
DELETE FROM EMPLOYEES WHERE E_ID IN (14,15,22,21);
```

```
SELECT * FROM OLD_EMPLOYEES;
```

```
SELECT * FROM EMPLOYEES_LOGS;
```